

MATH 252 – MATLAB ASSIGNMENT 01

due by Friday, June 3

Instructions

The purpose of this assignment is to introduce the student to basic tools available on MATLAB. The goal is to edit the file *Labs01_FormatFile.mlx* while attempting the task at hand. After you are done, doublecheck your answers and upload it to Canvas.

Exercises

- Basic computations: Set up variables $x = 6$ and $y = 7$, then compute
 - xy ,
 - $y^2 - y$
 - $(x + 2y)/0.5 + \sqrt{2x - y - 1}$
- Display of information: it is important to not only compute data, but to present it in a readable manner. For that, use the command *disp* to express the quantity below
 - x (answer provided as example)
 - y
 - xy
- If statements: Given a random number from 1 to 100, decide if it is even or not. Helpful commands are
 - $\text{randi}(100)$ will give you a random integer from 1 to 100,
 - $\text{mod}(a, m)$ returns the remainder after division of a by m
- For loops: a not-so appropriate (but valid) method to check if a number is prime or not is to check how many divisors it has. Write a for loop that check if 42 is a prime number by doing
 - Set up an empty vector $v = []$ to start. (answer provided as example)
 - Write a for loop going from 1 to 42 that does: if a number divides 42, it is added to v . You can do it by making $v(\text{end} + 1) = n$, where n is the number you decided to be a divisor of 42.
 - Display your results (answer provided as example)
- Plotting graphs: Draw a graph of the functions $f(x) = \sin(x)$ and $g(x) = x$ over the range $[-\pi, \pi]$. Draw f in blue and g in red.
 - Define the functions f, g
 - Draw f and g in separated figures.
 - Draw f and g in the same figure.

To make sure that figures separate, you can use the *figure()* command. In addition, *hold on* and *hold off* will make it so consecutive plots show up on the same axes. Use the function *plot()* to plot.

- Consider the differential equation

$$\frac{dy}{dt} = -2y + 2\cos(t)\sin(2t)$$

- What is the order of the diff. equation? Is it linear? Is it separable?
- Run the example code that plots different initial conditions for the differential equation. If a solution curve starts at $y(0)$ between $[-5, 5]$, what is its behavior as $t \rightarrow \infty$?